NON-AQUEOUS SOLVENTS

* Write the properties of solvent.

A solvent is the material present in soln in largen amount & is nesponsible fon dissolving the solute. The properties / cnitenia of a solvent anе –

1. The melting & boiling point of a solvent should be in the liquid nange.
2. It must have a dielectnic constant.
3. The solvent must have the ability of auto-ionization.
4. It should be able to fonm hydnogen bonding with ionic and polan solutes.
5. It should have e^- donon & acceptон pnopenties.
6. It should also have the ability to donate on accept pnotons.

* Define non-aqueous solvent with example.

A non-aqueous solvent is any solvent othen than watен which is used as a medium to dissolve solutes and is not an onganic compound. These solvents anе often chosen fon specific applications whenе watен is not suitable. Examples anе – Liquid ammonia, HF, punе H_2SO_4, BrF_5, N_2O_4, sulfunyl chlonide (SO_2Cl_2), phosphonyl chlonide ($POCl_3$) etc.

* Define protic and aprotic solvent.

Protic solvents are polar liquid compounds that have dissociable H-atoms, allowing them to form H-bonding. They can donate protons. Example : H_2O, HF, H_2SO_4, MeOH etc.

Aprotic solvents are liquid compounds that have no dissociable H-atoms and are less polar. They do not form H-bonding & cannot act as proton donors. Example : Dimethyl sulfoxide (DMSO), N_2O_4, BrF_3 etc.

* Explain the term 'leveling effect'.

Leveling effect refers to the phenomenon where the strength of a strong acid or a strong base is leveled when dissolved in a solvent.

For example : water exerts leveling effect on all strong acids & bases.

Water acts as a very strong base when reacted with strong acids. It has a great tendency to accept protons:

$$HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$$

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-$$

Thus, all strong acids react almost completely to form H_3O^+ ion. Therefore, all strong acids in water appear as equally strong ; i.e. acidic level for all acids are same. This is called leveling effect & water is the leveling solvent.

* Explain differentiating effect.

Differentiating effect is the effect of a solvent in which distinct acids dissociate to varying degrees & so have different strengths.

Weak bases are used as differentiating solvent. Glacial Acetic acid is a differentiating solvent. It can act as a very weak base by accepting a proton. Since it is a much weaker base than water, it is not as easily protonated & so there are appreciable differences in strength.

$$CH_3COOH + HClO_4 \rightleftharpoons CH_3COOH_2^+ + ClO_4^-$$

$$CH_3COOH + HBr \rightleftharpoons CH_3COOH_2^+ + Br^-$$

* Explain why CH_3COOH is a weak acid in H_2O but acts as a strong acid in NH_3?

CH_3COOH is a weak ~~be~~ acid in water while strong acid in NH_3. This is due to the nature of the solvents, i.e. H_2O & NH_3.

In water, acetic acid partially dissociates into ions –

$$CH_3COOH \xrightleftharpoons{H_2O} CH_3COO^- + H^+$$

Since this rxn does not go to completion & establishes an equilibrium, CH_3COOH is considered as a weak acid in water.

On the other hand, NH_3 is a ~~weaker~~ stronger base than water & has a higher

affinity for protons. When CH_3COOH is added to NH_3, it, thus, effectively accept protons from CH_3COOH to form ammonium ion. ~~Hence~~

$$CH_3COOH + NH_3 \rightleftharpoons CH_3COO^- + NH_4^+$$

Thus, NH_3 causes CH_3COOH to dissociate more than that of H_2O & so CH_3COOH acts as a strong acid in ammonia.

* Solvolysis is a chemical rxn in which the solvent, such as water, is one of the reagents and is present in excess amount than that required for the rxn.

$$CH_3Cl + 2NH_3 \longrightarrow CH_3NH_2 + NH_4Cl$$

* Explain why soln's of alkali metals in liq NH_3 are blue.

The soln's of alkali metals in liq. NH_3 are blue in colour.

When an alkali metal is dissolved in liq. NH_3, the metal donates an e^- to the soln. Both alkali metal & electrons become solvated by NH_3 molecules in soln.

$$M \xrightarrow{\text{dissociated in liq. } NH_3} M^+(solv) + e^-(solv)$$

The e^- that is released from the metal is called an ammoniated e^- as it is solvated in the soln. The blue colour of the soln arises from the presence of these ammoniated electrons. The energy levels of these electrons

allow them to absorb certain wavelengths of light, particularly in the visible spectrum, which results in the characteristic blue colour.

** The soln is highly conductive due to the presence of ~~both ammoniated cations & amo~~ ammoniated electrons. The electrons move freely in the soln contributing to the electrical conductivity of soln.

* How does liquid NH_3 enhance the acidity of compounds that are weak in water?

Certain acids that are weak in water, are strong in liquid NH_3.

Water is amphoteric in nature. It can accept a proton from a strong ~~base~~ acid & act as a base. ~~It has a higher dielectric constant (≈ 80), which stabilizes ions effectively, making it a strong ionizing solvent.~~

On the other hand, ammonia is a stronger base than water because NH_3 is a better proton acceptor, i.e, it exhibits higher affinity for protons. Hence the equilibrium of the rxn is shifted more to the right favouring the dissociation of acids more efficiently than water.

For example: CH_3COOH is a weak acid in water but acts as a strong acid in NH_3. This is because liq. NH_3 favours the dissociation of CH_3COOH more than that of water.

$$CH_3COOH \xrightleftharpoons{H_2O} CH_3COO^- + H_3O^+$$

$$CH_3COOH \xrightarrow[\longleftarrow]{\text{liq. } NH_3} CH_3COO^- + H_3O^+$$

Thus, liq. NH_3 enhanced the acidity of CH_3COOH.

** Soln's of metal in liq. NH_3 are paramagnetic in nature because — (আরো নতুন দেখায় বর্ণনা করতে হবে)

The dilute soln's ~~are p~~ contain free ammoniated electrons. These e^- make the soln paramagnetic. As the conc. is increased, extensive e^- pairing occurs which make the soln less paramagnetic.

** Soln's of metal in liq. NH_3 are good reducing agent — due to the presence of solvated and free electrons. They can reduce other compounds more efficiently, such as — $K_2[Ni(CN)_4]$ to $K_4[Ni(CN)_4]$.

* Show the auto-ionization of water & ammonia:

Both water and ammonia can undergo self-ionization.

Auto-ionization of NH_3:

The self-ionization of NH_3 involves 2 NH_3 molecules interacting to produce an ammonium ion (NH_4^+) & an amide ion (NH_2^-). One NH_3 molecule of the reactant acts as an acid & donates proton, while the other acts as a base & accepts the proton. The rxn is—

$$NH_3\,(i) + NH_3\,(ii) \rightleftharpoons NH_2^-\,(i) + NH_4^+\,(ii)$$

acid — base — conjugated base — conjugated acid

The self-ionization constant for NH_3 ~~(k)~~ is much smaller than that of water due to weaker ionizing nature of liq. NH_3.

Auto-ionization of water:

In this process, two water molecules react to give a hydronium ion (H_3O^+) & a hydroxide ion (OH^-).

The rxn is —

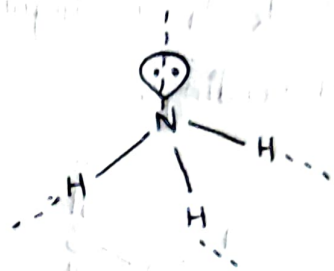
$$H_2O\,(i) + H_2O\,(ii) \rightleftharpoons H_3O^+\,(i) + OH^-\,(ii)$$

base — acid — conj. acid — conj. base.

* Why is the boiling point of water higher than ammonia?

The boiling point of H_2O is higher than NH_3. It is 100°C for H_2O but -33°C for NH_3.





Both H_2O & NH_3 have simple molecular structure & their molecules are held together by H-bonds. From the diagrams, we see that each H_2O molecule has 2 lone pairs of e^- & 2 H-atoms to form H-bonds. Averagely, there are 2H-bonds per water molecule.

While, each NH_3 molecule has 1 lone pair of e^- & 3 H-atoms. Averagely there is 1 H-bond per NH_3 molecule.

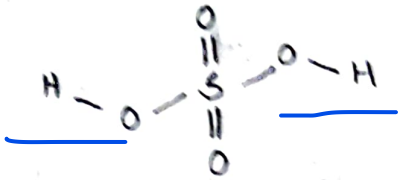
Since water has more H-bonds per molecule, more energy is needed to seperate water molecules than NH_3 molecules. Hence the b.p of H_2O is higher than that of NH_3.

* Why is NH_3 a better solvent than water?

1. Ammonia can form less H-bonds than water, which allows it to dissolve a wider range of substances.
2. NH_3 ~~remo~~ has a wider range of liquid stability ~~at~~ under pressure than water.
3. It is less viscous than water.

* Explain that anhydrous H_2SO_4 is highly viscous & has low ion mobilities.

Anhydrous H_2SO_4 is highly viscous & has low ion mobility due to H-bonding.



Each H_2SO_4 molecule has two highly polar hydroxyl (-OH) groups & a strongly electronegative O-atoms for which it can form strong H-bonding with its molecules. The presence of H-bonds creates strong attractions betⁿ adjacent molecules resulting in a higher viscosity. Again, the presence of multiple H-bonds creates a dense, interconnected structure of H_2SO_4, which causes significant resistance to the movement of ions. Thus, this high viscosity of anhydrous H_2SO_4 slows down the transport of ions through the medium. Therefore, anhydrous H_2SO_4 is highly viscous & has low ion mobilities.

* Compare the solvent properties of Liq NH_3 & H_2O :

1) Liquid range : The m.p & b.p. of liq NH_3 are lower than that of water.

Liquid range of liq NH_3 : $-77^\circ C$ to $-33^\circ C$

Liquid range of ~~Ro~~ H_2O : $0^\circ C$ to $100^\circ C$

2) Dielectric constant : The dielectric constant of liq. NH_3 is lower comparing to water which makes Liq NH_3 a poor solvent for ionic substances.

3) H-bonding : The number of H-bonding in per ~~liq~~ NH_3 molecule is ~~lowe~~ smaller than that of water.

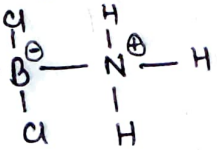
4) Viscosity : Ammonia is less viscous than water.

5) Self-ionization : The extent of self-ionization of liq. NH_3 is less than that of water. Hence, liq. NH_3 has lower specific conductance than water.

* Complete the rxns -

1) $KI + NH_4Cl \xrightarrow{\text{liq } NH_3} KCl\downarrow + NH_4I$

2) $NH_4Cl + KNH_2 \xrightarrow{\text{liq } NH_3} 2NH_3 + KCl$

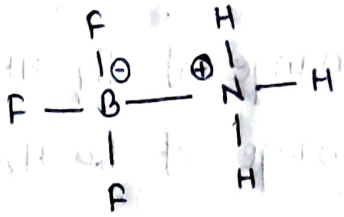
3) $BCl_3 + NH_3 \xrightarrow{\text{liq } NH_3}$ 

4) $2FeCl_3 + 2KI \xrightarrow{\text{liq } SO_2} 2FeCl_2 + 2KCl + I_2$

5) $SbCl_3 + 3KCl \xrightarrow{\text{liq } SO_2} K_3[SbCl_6]$

6) $3\,SbCl_5 + 6\,KI \xrightarrow{liq\ SO_2} 3\,SbCl_3 + 6\,KCl + 3\,I_2$

7) $2\,KI + SOCl_2 \xrightarrow{liq\ SO_2} 2\,KCl\downarrow + SO\,I_2$

8) $BF_3 + NH_3 \xrightarrow{liq\ NH_3}$ 

9) $PH_3 + Na \xrightarrow{liq\ NH_3} Na\,PH_2 + \frac{1}{2}H_2$

10) $CH_3COOH + H_2SO_4 \xrightarrow{liq\ H_2SO_4} [CH_3C(OH)_2]^{+} + HSO_4^{\ominus}$

11) $NH_2CONH_2 + H_2SO_4 \xrightarrow{liq\ H_2SO_4} NH_2CO\,NH_3^{\oplus} + HSO_4^{\ominus}$